

Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE-4040 – Digital Electronics
Spring Term 2026

Lab. 7- Delays and Oscillator Circuits

Issued: 3/11/2025

Due: 3/13/2026

Given

Below is a table of process parameters for a generic 0.13 μm technology.

0.13 μm Process technology parameters

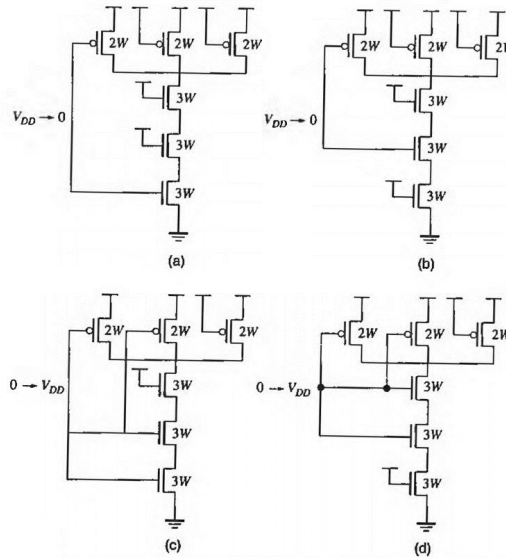
Typical Technology Parameters	Symbol	NMOS	PMOS
Gate thickness	t_{ox} [Å]	22	22
Poly doping level	N_D [cm^{-3}]	3×10^{20}	—
Poly doping level	N_A [cm^{-3}]	—	3×10^{20}
Substrate doping level	N_A [cm^{-3}]	3×10^{17}	—
N-well doping level	N_D [cm^{-3}]	—	3×10^{17}
Number of surface state charges	N_{ss} [cm^{-2}]	6×10^{11}	6×10^{11}
Electron Mobility	μ_n [cm^2/Vs]	270	—
Hole Mobility	μ_p [cm^2/Vs]	—	70
Critical Field x Length	$E_C L$ [V]	0.6	2.4
Saturation velocity	v_{sat} [cm/s]	8×10^6	8×10^6
Gamma	γ [$\text{V}^{1/2}$]	0.2	0.2
Oxide capacitance	C_{ox} [F/cm^2]	1.6×10^{-6}	1.6×10^{-6}
Threshold voltage	V_T [V]	0.4	− 0.4
2 x Fermi voltage	$2 \phi_F $ [V]	0.88	0.88
Channel length modulation	λ [V^{-1}]	0.2	0.2

Question 1: Based on Problem 6.4 (D.A. Hodges, H.G. Jackson, and R.A. Saleh, *Analysis and Design of Digital Integrated Circuits*, 3rd Ed., McGraw-Hill, 2004.)

Objectives:

- Logic gate Elmore delay calculation

Given the NAND3 gate below at various input change scenarios on 0.13 μ m process:



- In the four input change scenarios shown above, calculate the $\tau_P = RC$ delay when only self (output) capacitance is included. Assume no source/drain sharing.
 - Recalculate the NAND3 output delay when a $W_P/L = 2$, $W_N/L = 1$ CMOS Inverter is added to the output (this makes the AND function).
 - In addition to all previously included capacitances, recalculate the NAND3 output delay when a 100 μ m long interconnect separates the NAND3 output and the INV input.
- Simulate the case with just C_{self} for all 4 input scenarios in SPICE.

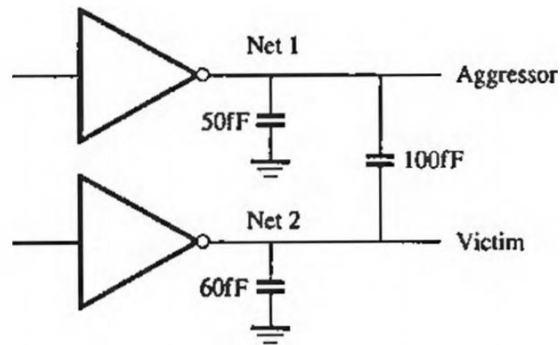
HINT:

- Convert each scenario into a RC chain, as typically seen in Elmore delay problems.
 - Each transistor output will have a node C, correlated to the transistor width.
 - Each transistor that is on will be converted to a resistor, with $R_{eff} = R_N * L/W$, or $R_{eff} = R_P * L/W$, with $R_N = 12.5k\Omega$, and $R_P = 30k\Omega$.
 - Each transistor that is off will be replaced with an open circuit.
- Now solve for the τ_P between the output node and V_{DD} (if output = 1) or GND (if output = 0) using Elmore delay analysis.

Question 2: Based on Problem 10.8 (D.A. Hodges, H.G. Jackson, and R.A. Saleh, *Analysis and Design of Digital Integrated Circuits*, 3rd Ed., McGraw-Hill, 2004.)

Objectives:

- Calculate noise voltage from capacitive coupling.
- Calculate increase in delay from capacitive coupling.



Given: $V_{DD} = 1.2V$, ignore Inverter self (output) capacitance.

- Estimate the peak value of noise on the victim line for the following scenarios:
 - Aggressor switches from logic 0 $\rightarrow$ 1 when the victim is logic 0.
 - Aggressor switches from logic 1 $\rightarrow$ 0 when victim is logic 1.
- What is the effective capacitance loading on the victim for delay calculation if the aggressor switches in the opposite direction of the victim?
 - What is the increase in % of delay when the aggressor switches in the opposite direction of the victim, versus when both aggressor and victim change in the same direction?

Report Submission:

Submit your work on LMS under **Week 8/Lab 7**.

- Show your work.
- Include images of your hand calculations with work and final results.
- Include screenshots of LTspice with all circuit schematics with all output plots, **clearly showing all hand calculated parameters**.
- Due date: Friday at 11:59pm.

